



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1303 – Probability Theory
Tutorial No: 3

1. A forester studying the effects of fertilization on certain pine forests in the Southeast is interested in estimating the average basal area of pine trees. In studying basal areas of similar trees for many years, he has discovered that these measurements (in square inches) are normally distributed with standard deviation approximately 4 square inches. If the forester samples $n = 9$ trees, find the probability that the sample mean will be within 2 square inches of the population mean.
2. Ammeters produced by a manufacturer are marketed under the specification that the standard deviation of gauge readings is no larger than .2 amp. One of these ammeters was used to make ten independent readings on a test circuit with constant current. If the sample variance of these ten measurements is .065 and it is reasonable to assume that the readings are normally distributed, do the results suggest that the ammeter used does not meet the marketing specifications? [Hint: Find the approximate probability that the sample variance will exceed .065 if the true population variance is .04.]
3. Is there difference between the variances of the number of weeks on the best seller list for the notification and fiction books? Fifteen New York Times bestselling fiction books had a standard deviation of 6.17 weeks on the list. Sixteen New York times bestselling notification books had a standard deviation of 13.12 weeks. at the 10% significance level, can we conclude there is a difference in the variances?
4. Two additives to Portland cement are being tested for their effect on the strength of concrete. 21 batches were made with Additive A, and their strengths showed standard deviation $S_A = 41.3$. 16 batches were made with the same percentage of Additive B, and their strengths showed standard deviation $S_B = 26.2$. Assume that the strengths of concrete follow a normal distribution. Is there evidence at the 1% level of significance that the concrete made with Additive A is more variable than concrete made with Additive B?
5. The tensile strength for a type of wire is normally distributed with unknown mean μ and unknown variance σ^2 . Six pieces of wire were randomly selected from a large roll; Y_i , the tensile strength for portion i , is measured for $i = 1, 2, \dots, 6$. The population mean μ and variance σ^2 can be estimated by $\bar{Y}$ and S^2 , respectively. Because $\sigma_y^2 = \frac{\sigma^2}{n}$,

i follows that σ_y^2 can be estimated by $\frac{s^2}{n}$. Find the approximate probability that Y will be within $\frac{2s}{\sqrt{n}}$ of the true population mean μ .

6. Researchers collected a simple random sample of the times that 81 college students required to earn their Bachelor's degrees. The sample has a mean of 4.8 years and standard deviation of 2.2 years.
 - a. Use a 0.05 significance level to test the claim the mean time for all college students is greater than 4.5 years.
 - b. Use a 0.05 significance level to test the claim the mean time for all college students is different from 4.5.

(Please answer all the questions and submit the answer script to the tutorial room on or 25th of October 2023)